

OCI Cloud Architect — Important Topics

Mid-Semester Exam Preparation · Day 1 – Day 6 Coverage

Subject Code: 303105518 · Faculty of Engineering & Technology, Parul University

The topics below are filtered strictly to items that appear in the question paper **and** are explicitly listed in the Day 1–6 planner. Every bullet is exam-relevant.

Day 1 · Cloud Foundations & IAM

Focus: OCI global infrastructure, resiliency, identity, compartments

- **Regions, Availability Domains, Fault Domains** — what each protects against
- **Shared Responsibility Model** — provider vs customer duties
- **Compartments & Tagging Strategy**
- **Identity Domains, Users & Groups, IAM Policies, Federation & SSO**

Day 2 · Security & Cost Governance

Focus: Security posture services, cost management, governance

- **CIA Triad**
- **Defense-in-Depth** — with an OCI example
- **Zero Trust** — vs traditional perimeter model
- **Cloud Guard, Security Zones, Security Advisor** — what each does; what Security Zones enforces that Cloud Guard doesn't
- **Organization Management**
- **Budgets & Alerts, Cost Analysis**
- **BYOL vs License Included, Reserved Capacity, OCI Marketplace, Cloud Advisor** — cost-reduction levers
- **Compliance Readiness & Governance Frameworks**

Day 3 · Linux, Networking & CLI

Focus: Linux fundamentals, networking basics, OCI CLI

- **Linux Permissions & chmod** — octal notation, owner/group/other
- **IPv4 & CIDR, Subnetting** — total / reserved / usable addresses
- **TLS/SSL & SSH** — timeout vs Permission denied (publickey) diagnostics
- **Load Balancer Basics** — 502 causes, backend set / listener config
- **OCI Cloud Shell & OCI CLI** — config file ~/.oci/config, auth differences

Day 4 · Infrastructure as Code

Focus: Terraform + OCI Resource Manager

- **IaC — Declarative vs Imperative, idempotence**
- **Terraform Workflow** — init, plan, apply, destroy
- **Terraform State & State Locking** — why remote state; concurrent-apply risk
- **OCI Resource Manager & Stacks** — advantages over laptop Terraform

- **Drift Detection**
- **Resource Discovery** — generating TF config + state from existing resources

Day 5 · VCN, Compute & Storage

Focus: Networking deep dive, compute, storage

- **VCN Architecture — Public/Private Subnets, Route Tables** — three-tier design
- **Internet Gateway, NAT Gateway, Service Gateway** — purpose of each; which for a private DB subnet
- **NSGs & Security Lists**
- **Compute Instances & Cloud-init** — cloud-config for first-boot web server
- **Boot Volume vs Block Volume** — multi-attach + filesystem requirement
- **SSH Key Injection**

Day 6 · Encryption & Key Management

Focus: OCI Vault, KMS, key protection

- **OCI Vault & KMS**
- **Master Encryption Key vs Data Encryption Key**
- **Key Protection Modes** — Software vs HSM

Weightage Snapshot

Day	Focus Area	Exam Weight
Day 1	Cloud Foundations & IAM	Medium
Day 2	Security & Cost Governance	High
Day 3	Linux, Networking & CLI	Medium
Day 4	Infrastructure as Code	High
Day 5	VCN, Compute & Storage	High
Day 6	Encryption & Key Management	Low

Study tip: Prioritise Day 2, Day 4, and Day 5 — these carry the most marks in the paper. Day 1 shows up mainly through the tenancy operating-model question. Day 6 has a single direct question but is quick to prepare.